

THERMODYNAMIC FORMALISM MEETS ERGODIC OPTIMIZATION

JUAN RIVERA-LETELIER

This note presents a concrete example illustrating the following connection between ergodic optimization and the thermodynamic formalism, in the context of MANNEVILLE–POMEAU maps and HÖLDER continuous potentials:

(*) *If a potential undergoes a phase transition in temperature, then the DIRAC measure at the neutral fixed point uniquely maximizes the potential.*

This result is a direct consequence of [CIR25, Key Lemma and Proposition 2.10] and the concrete example appears as [CIR25, Example 1.2]. For background on ergodic optimization, see, for example, [Jen19].

The example relies on the following notation and terminology. Fix α in $(0, +\infty)$ and the MANNEVILLE–POMEAU or *intermittent* map $f: [0, 1] \rightarrow [0, 1]$ defined by

$$(1) \quad f(x) := \begin{cases} x(1 + x^\alpha) & \text{if } x(1 + x^\alpha) \leq 1; \\ x(1 + x^\alpha) - 1 & \text{otherwise.} \end{cases}$$

Denote by $\mathcal{M}$ the space of BOREL probability measures on $[0, 1]$ that are invariant by f and, for each ν in $\mathcal{M}$, by h_ν the measure-theoretic entropy of ν . For each continuous function $\varphi: [0, 1] \rightarrow \mathbb{R}$, the *pressure* $P(\varphi)$ of f for the potential φ is defined by

$$(2) \quad P(\varphi) := \sup \left\{ h_\nu + \int \varphi \, d\nu : \nu \in \mathcal{M} \right\}.$$

Proposition 1. *Let γ in $(0, +\infty)$ be given, and define the potential $\hat{\varphi}$ by*

$$(3) \quad \hat{\varphi}(x) := -x^\gamma(1 - x).$$

Then, $t \mapsto P(t\hat{\varphi})$ is real-analytic on $(0, +\infty)$.

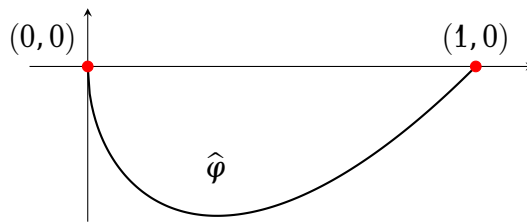


FIGURE 1. The graph of the potential $\hat{\varphi}$ with $\gamma = \frac{1}{2}$.

The potential $\hat{\varphi}$ is nonpositive and vanishes precisely at the neutral fixed point 0 and at the repelling fixed point 1. Thus, the measure δ_0 maximizes $\hat{\varphi}$, but not uniquely. The conclusion of **Proposition 1** is that the potential $\hat{\varphi}$ does not undergo a phase transition in temperature, as ensured by (*).

The remainder of this note proves [Proposition 1](#). The proof relies on a couple of lemmas. Denote by x_1 the unique discontinuity point of f .

Lemma 2. For every β in $(0, +\infty)$,

$$(4) \quad \sup_{[0,1]} \beta \hat{\varphi} < P(\beta \hat{\varphi}).$$

In particular, the potential $\beta \hat{\varphi}$ is hyperbolic for f in the sense of [\[IRRL12\]](#).

Proof. Denote by x_2 the unique preimage of x_1 by f in $(0, x_1)$, and by X the maximal invariant set of f in $[x_2, 1]$. Then, $f|_X$ is uniformly expanding and topologically conjugate to the golden mean shift, via a HÖLDER continuous conjugacy. By the thermodynamic formalism for HÖLDER continuous potentials, it follows that $f|_X$ has a unique GIBBS state for the potential $\beta \hat{\varphi}|_X$ and that the measure-theoretic entropy of this measure is strictly positive. See, for example, [\[Bow08, Theorem 1.25\]](#). It follows that δ_1 is not an equilibrium state of $f|_X$ for the potential $\beta \hat{\varphi}|_X$, and thus

$$(5) \quad P(\beta \hat{\varphi}) \geq P_{f|_X}(\beta \hat{\varphi}|_X) > h_{\delta_1}(f) + \int \beta \hat{\varphi}|_X d\delta_1 = 0.$$

Since $\hat{\varphi} \leq 0$ on $[0, 1]$, the inequality [\(4\)](#) follows. $\square$

Lemma 3. Let β be in $(0, +\infty)$ and put $g := \exp(\beta \hat{\varphi} - P(\beta \hat{\varphi}))$. Then, there is a $(1/g)$ -conformal probability measure m for f that is atom-free and satisfies $\text{supp}(m) = [0, 1]$.

Proof. Put $I := [0, 1]$ and let $\tilde{f}: \tilde{I} \rightarrow \tilde{I}$ be the continuous extension of f obtained by “doubling” the points of $\bigcup_{n=0}^{\infty} f^{-n}(x_1)$, as in [\[CIR25, Proof of Lemma 2.6\]](#). Put $Y^+ := \bigcup_{n=0}^{+\infty} \tilde{f}^{-n}(x_1^+)$, and denote by $\pi: \tilde{I} \rightarrow I$ the corresponding projection and $\tilde{\varphi}: \tilde{I} \rightarrow \mathbb{R}$ the continuous extension of $\hat{\varphi}|_{\tilde{I} \setminus Y^+}$. The map $\tilde{f}$ is a local monotone homeomorphism that is topologically exact. Furthermore, $\pi \circ \tilde{f} = f \circ \pi$, $\tilde{\varphi} = \hat{\varphi} \circ \pi$, and $\pi: \tilde{I} \setminus Y^+ \rightarrow I$ is a BOREL measurable bijection conjugating $\tilde{f}|_{\tilde{I} \setminus Y^+}$ to f .

Since $\tilde{f}$ is topologically conjugate to the full shift in two symbols [\[CIR25, Lemma 2.6\]](#), the dual $\mathcal{L}_{\tilde{\varphi}}^*$ of the transfer operator $\mathcal{L}_{\tilde{\varphi}}$ of $\tilde{f}$ for the potential $\tilde{\varphi}$ is thus continuous and has an eigenprobability $\tilde{m}$ by the SCHAUDER–TYCHONOFF Fixed Point Theorem. Using, for every n in $\mathbb{N}$, the equality $(\mathcal{L}_{\tilde{\varphi}}^*)^n \tilde{m}(\tilde{I}) = \int \mathcal{L}_{\tilde{\varphi}}^n(\mathbf{1}_{\tilde{I}}) d\tilde{m}$ and that $\frac{1}{n} \log \mathcal{L}_{\tilde{\varphi}}^n \mathbf{1}_{\tilde{I}}$ converges pointwise to $P_{\tilde{f}}(\tilde{\varphi})$ as $n \rightarrow +\infty$ [\[CIR25, Proposition 2.9\(2\)\]](#), it follows that $\mathcal{L}_{\tilde{\varphi}}^* \tilde{m} = \exp(P_{\tilde{f}}(\tilde{\varphi})) \tilde{m}$ and $\tilde{m}$ is $\exp(\beta \tilde{\varphi} - P(\beta \tilde{\varphi}))$ -conformal for $\tilde{f}$. The equality $\text{supp}(\tilde{m}) = \tilde{I}$ follows from the fact that $\tilde{f}$ is topologically exact on $(0, 1]$ [\[CIR25, Lemma 2.8\]](#).

To prove that $\tilde{m}$ is atom-free, let $\tilde{x}$ in $\tilde{I}$ be given. For every k in $\mathbb{N}$,

$$(6) \quad \begin{aligned} \tilde{m}(\{x\}) &= \exp(-(kP(\hat{\varphi}) - S_k(\tilde{\varphi})(\tilde{x}))) \tilde{m}(\{\tilde{f}^k(x)\}) \\ &\leq \exp\left(-k \left(P(\hat{\varphi}) - \sup_{[0,1]} \hat{\varphi}\right)\right) \tilde{m}(\{\tilde{f}^k(\tilde{x})\}). \end{aligned}$$

Letting $k \rightarrow +\infty$ and using [\(4\)](#) in [Lemma 2](#) yields $\tilde{m}(\{\tilde{x}\}) = 0$. This proves that $\tilde{m}$ is atom-free.

Put $m := \pi_* \tilde{m}$. Since $\text{supp}(\tilde{m}) = \tilde{I}$ and $\tilde{m}$ is atom-free, $\text{supp}(m) = I$ and m is atom-free and $(1/g)$ -conformal for f . $\square$

Proof of Proposition 1. Let β be in $(0, +\infty)$, put $g := \exp(P(\beta\hat{\varphi}) - \beta\hat{\varphi})$ as in Lemma 3, and put $\gamma_* := \min\{1, \gamma\}$. The functions $\hat{\varphi}$ and g belong to $C^{\gamma_*}([0, 1], \mathbb{R})$ and g satisfies $\sup_{[0, 1]} g < 1$ by Lemma 2. Let m be the atom-free $(1/g)$ -conformal probability measure for f satisfying $\text{supp}(m) = [0, 1]$ given by Lemma 3.

The first step is to verify the hypotheses of [LRL14, Theorem 1 in §4.2], which is a reformulation of results of KELLER [Kel85, Theorems 3.2 and 3.3]. Hypotheses H1 and H2 of [LRL14, Theorem 1 in §4.2] follow automatically from the fact that every function in $C^{\gamma_*}([0, 1], \mathbb{R})$ is of bounded $(1/\gamma_*)$ -variation and that m is $(1/g)$ -conformal for f . Hypothesis H3 of [LRL14, Theorem 1 in §4.2] can be shown as in the proof of [LRL14, Lemma 5.1]. Thus, the hypotheses of [LRL14, Theorem 1 in §4.2] are satisfied. Using that f is topologically exact on $(0, 1]$ [CIR25, Lemma 2.8], [LRL14, Corollary 4.4(1)] yields that $t \mapsto P(t\varphi)$ is real-analytic at $t = \beta$. $\square$

REFERENCES

- [Bow08] Rufus Bowen. *Equilibrium states and the ergodic theory of Anosov diffeomorphisms*, volume 470 of *Lecture Notes in Mathematics*. Springer-Verlag, Berlin, revised edition, 2008. With a preface by David Ruelle, Edited by Jean-René Chazottes.
- [CIR25] Daniel Coronel, Irene Inoquio-Renteria, and Juan Rivera-Letelier. Phase transitions in temperature for intermittent maps. *arXiv e-prints*, page arXiv:2504.01166, April 2025.
- [IRRL12] Irene Inoquio-Renteria and Juan Rivera-Letelier. A characterization of hyperbolic potentials of rational maps. *Bull. Braz. Math. Soc. (N.S.)*, 43(1):99–127, 2012.
- [Jen19] Oliver Jenkinson. Ergodic optimization in dynamical systems. *Ergodic Theory Dynam. Systems*, 39(10):2593–2618, 2019.
- [Kel85] Gerhard Keller. Generalized bounded variation and applications to piecewise monotonic transformations. *Z. Wahrsch. Verw. Gebiete*, 69(3):461–478, 1985.
- [LRL14] Huaibin Li and Juan Rivera-Letelier. Equilibrium states of interval maps for hyperbolic potentials. *Nonlinearity*, 27(8):1779–1804, 2014.

DEPARTMENT OF MATHEMATICS, UNIVERSITY OF ROCHESTER. Hylan Building, ROCHESTER, NY 14627, U.S.A.
Email address: riveraletelier@gmail.com
URL: <http://rivera-letelier.org/>